

Master Face Attacks on Facial Recognition Systems: A Comprehensive Review

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May 23, 2026

Abstract

This literature review examines current research surrounding biometric authentication using facial recognition systems (FRS), focusing on the feasibility of generating "Master Faces" and the resilience of state-of-the-art (SOTA) systems against them. Master Faces, or "wolf samples," are synthetic identities optimized via Latent Variable Evolution (LVE) to exploit "dense areas" in biometric embedding spaces where identity separability is poor. Utilizing a systematic literature review methodology aligned with PRISMA guidelines, we analyze the evolution of these attacks from early demonstrations—which reported that fewer than ten Master Faces could successfully match over 40% of identities in legacy datasets like LFW to recent empirical findings. Our results reveal a significant "generalizability gap": while Master Faces are potent against older, low-dimensional models like FaceNet or Dlib, their success rates drop significantly to between 0.01% and 0.31% when tested against modern high-dimensional models ($d \geq 128$) utilizing angular margin losses, such as ArcFace, MagFace, and CurricularFace. This study identifies that Master Faces act as "model-specific keys" rather than universal keys, with limited transferability to unseen architectures. We conclude that while modern architectural hardening is inherently resilient, robust protection in enterprise security requires a layered system integrating multi-modal fusion (e.g., face, voice, and fingerprint) and action-based liveness detection.

Keywords

Adversarial attacks, Face recognition systems (FRS), Master Face attack, Convolutional Neural Network (CNN), Deep Neural Network (DNN)

1 Introduction

The rapid global adoption of Facial Recognition Systems (FRS) as a primary modality for biometric authentication has fundamentally reshaped security infrastructures in sectors ranging from mobile payment and e-banking to automated border control and enterprise access management [1][2][3]. As FRS becomes more deeply enmeshed in everyday digital life, it has emerged as a high-value target for threat actors seeking to exploit vulnerabilities and bypass Multi-Factor Authentication (MFA) frameworks [4][5]. Among the most sophisticated and evolving of these threats is the Master Face attack, the creation of a synthetic "master key" or "wolf sample" designed to be falsely accepted as a match against a disproportionately large portion of a given population without requiring any prior knowledge of the enrolled subjects [6][7][8][9].

The core technical challenge posed by Master Face attacks lies in their exploitation of the underlying distribution of biometric embedding spaces [8]. Modern FRS typically use Deep Neural Networks (DNNs) to map facial images into high-dimensional feature vectors [10]. Master Faces, often generated using Latent Variable Evolution (LVE) algorithms paired with high-resolution generators like StyleGAN, are optimized to find "dense areas" or regions of poor identity separability within these embedding spaces [8][9][11]. By occupying these regions, a single synthetic face can theoretically compromise a significant percentage of a database. For example, early seminal research reported that a small set of fewer than ten Master Faces could successfully cover over 40% of identities in benchmark datasets such as LFW when tested against older models such as FaceNet or Dlib [6][8].

However, the current literature reveals a critical generalizability gap regarding the actual

threat level posed to modern, SOTA systems [7][11]. While proof-of-concept attacks have demonstrated high success rates in white-box scenarios where the attacker has full access to the target model, more recent comprehensive evaluations suggest these findings may be artifacts of over-optimization on low-dimensional representations [11][7]. When tested against modern high-dimensional models utilizing angular margin losses (e.g., ArcFace, MagFace, CurricularFace), the effectiveness of these attacks often drops to levels comparable to "zero-effort" imposter attacks, with success rates falling as low as 0.01% to 0.31% [8][11]. This discrepancy raises significant questions about whether Master Faces are truly "universal keys" or merely "model-specific keys" whose utility vanishes when the target architecture or training distribution changes [11].

Furthermore, there is a lack of consensus on the most effective mitigation and defense strategies [8]. Current research is divided between architectural defenses, such as balancing training data across demographics to eliminate vulnerable clusters, and procedural defenses, which involve integrating dedicated GAN-based image detectors or enhanced liveness detection wrappers to filter synthetic artifacts [8]. A key unresolved gap remains the digital-to-physical transfer loss, as researchers disagree on how a Master Face survives a complete real-world pipeline that includes simultaneous face detection, quality assessment, and action-based liveness checks [8][12][13][14].

The objective of this study is to systematically evaluate the vulnerability of modern facial recognition models to Master Face attacks, assessing their resilience in the context of diverse SOTA architectures and real-world environmental variations. By analyzing the conflicts between theoretical capacity and empirical performance, this research aims to identify the optimal model implementation and defense configurations capable of preventing the Master Face threat, thus fortifying the integrity of biometric authentication in high-stakes enterprise security environments [15][8].

2 Background & Significance

2.1 Deep Neural Networks (DNN)

The creation of modern biometric models relies on the development of Deep Neural Networks (DNNs), which serve as the foundational architecture for mapping raw facial data into high-dimensional feature vectors (also known as templates or embeddings) [7][8][16]. These networks are created through a process of inductive representation learning, where a mapping function

f is optimized using a loss function to ensure that images of the same identity are clustered closely together in the identity space while different identities are pushed apart [8][16]. The training of these models is extremely data-intensive, requiring large-scale datasets such as CASIA-WebFace or MS-Celeb-1M and massive computational resources, often provided via Machine Learning-as-a-Service (MLaaS) platforms [17][18]. SOTA DNNs utilize multiple layers of processing units to learn abstract representations that demonstrate invariance against environmental variations like lighting, pose, and expression [3].

2.2 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are a specialized subclass of DNNs designed specifically for computer vision tasks such as face detection, alignment, and feature extraction [7][19]. A CNN is structured into three primary types of layers:

- **Convolutional Layers:** These layers use filter kernels to execute nonlinear transfer operations, extracting essential spatial features and edge definitions from the 2D input image [20][21][16].
- **Pooling Layers:** These functions down-sample the feature maps by taking the maximum or average value of adjacent pixel patches, which minimizes the area of the maps and reduces computational load [20][16].
- **Fully Connected (FC) Layers:** Located at the end of the network, these layers perform the final classification or map the extracted features to a lower-dimensional real-valued vector for similarity comparison [20][16][21].

Modern CNNs typically leverage established "backbone" architectures such as ResNet, VGG-Net, or Inception, which are often pre-trained on large-scale datasets and then fine-tuned for specific biometric authentication tasks [16][7][22].

2.3 Model Vulnerabilities and the Master Face Attack

The inherent structure and training objectives of DNNs and CNNs provide the mathematical "singularities" that allow a Master Face attack to occur [23]. While these models attempt to create distinct identity boundaries, the embedding space they create is often not uniformly distributed, resulting in "dense areas" where feature extractors fail to achieve perfect identity separability [8]. A Master Face (or "wolf sample") is

a synthetic identity optimized to occupy a specific point in these dense areas, allowing it to be falsely accepted as a match for a disproportionately large portion of the enrolled population [8]. The attack exploits the fact that neural networks are approximations rather than exact matches; thus, a single "master key" can lie within the decision threshold of multiple distinct identity clusters [8]. Furthermore, because advanced models are often over-optimized on specific training distributions, they develop model-specific "wolf regions" that an attacker can reverse-engineer using Latent Variable Evolution (LVE) to create highly effective, though often model-specific, impersonation images [8][11].

3 Related Work

The landscape of adversarial research in facial recognition systems (FRS) is characterized by a rapid evolution of attack vectors and a corresponding struggle to establish generalized defense frameworks. Prior exemplar literature reviews have categorized these threats into distinct taxonomies, primarily distinguishing between presentation attacks (physical artifacts) and template/model group attacks (digital manipulations) [2]. While comprehensive surveys have addressed general biometric vulnerabilities, the specific sub-topic of Master Face attacks remains an area of intense debate regarding its practical generalizability [11][8].

3.1 Taxonomy and Structural Benchmarks

Existing reviews provide the structural foundation for understanding Master Face attacks within the broader FRS security ecosystem. Leyva et al. present a systematic review of attack vectors from 2017-2024, emphasizing that while presentation attacks are the most frequent due to low barriers to entry, template-level attacks are becoming increasingly sophisticated and severe [2]. This work establishes an Architecture Reference Architecture (ARA) to identify specific points of failure in the FRS pipeline, a method we adopt to analyze the "wolf regions" in biometric embedding spaces [2][8]. Furthermore, Vakhshiteh et al. and Zhang et al. provide extensive studies on adversarial machine learning, noting that the field is shifting from "zero-effort" imposter models to active synthetic impersonation [24][5][25]. These reviews agree that modern FRS, though achieving near-perfect accuracy on standard datasets like LFW, remain vulnerable to Latent Variable Evolution (LVE) techniques used to create master keys [6][23]. How-

ever, they also highlight a significant gap: most prior studies rely on older, low-dimensional models (e.g., FaceNet, Dlib), leaving the resilience of high-dimensional SOTA systems under-explored [11].

3.2 Synthesizing Generation and Mitigation Strategies

The methodology for generating Master Faces via StyleGAN2 and evolutionary algorithms is well-documented in seminal works [8][6][9]. Venkatesh et al. provide a comprehensive survey on face morphing, which shares technical commonalities with Master Face generation but differs in its requirement for specific victim data [8][26]. These reviews reveal a consensus that Master Faces exploit "dense areas" of poor identity separability in the embedding space [8]. Disagreement persists, however, regarding mitigation. Some researchers, as noted by Khairnar et al., advocate for procedural defenses like specialized GAN-detectors to filter synthetic artifacts [1][8]. Conversely, other literature focuses on architectural fixes, such as training with Angular Margin Loss (ArcFace) or Uniform Loss to ensure even embedding distributions [8][16]. The conflict between these strategies suggests that the field has yet to "crown" an optimal implementation that maintains recognition accuracy while preventing population-matching synthetic identities.

3.3 Core Question and Sub-Topic Gaps

Despite these reviews, a CORE QUESTION remains: *To what extent does the success of a Master Face attack depend on the specific training distribution of the target model versus the mathematical properties of high-dimensional embedding spaces?* Current literature is divided; early studies report success rates over 40%, while recent cross-model evaluations show rates as low as 0.31% [6][11]. A SUB-QUESTION asks: *How can a literature review identify the specific architectural "sweet spot" (in terms of dimensionality and loss function) where Master Face coverage becomes statistically insignificant?* By synthesizing empirical results from diverse sources that test across varying dimensionalities ($d=128$ vs. $d=512$) and loss functions (ArcFace vs. MagFace), this study aims to fill the generalizability gap identified across the literature [11][9]. We move beyond isolated proof-of-concepts to evaluate the resilience of the entire end-to-end pipeline, which exemplar reviews often neglect in favor of isolated matching stage analysis [14][19].

4 Research Method

This study is a systematic literature review designed to analyze the evolution of facial recognition vulnerabilities by synthesizing findings from across the technical landscape to address the generalizability gap in Master Face attack research [27][28][29]. Utilizing a methodology aligned with established PRISMA guidelines, the review systematically indexes and evaluates high-impact research from primary databases, including IEEE Xplore, ACM Digital Library, and ScienceDirect, to ensure a comprehensive and unbiased categorization of threat vectors and defense mechanisms [1][2][4]. By critically assessing SOTA models, diverse datasets, and the impact of angular margin losses, this research aims to provide an objective framework for identifying optimal implementation strategies to mitigate population-matching synthetic identity threats and bolster the integrity of biometric authentication [2][7].

5 Findings

5.1 Technical Analysis

The Master Face attack represents a high success form of a biometric dictionary attack, often referred to as a wolf attack [8][11]. Unlike face morphing, which typically requires access to a specific victim’s photograph, a Master Face is a synthetic image generated to match a large percentage of a facial recognition system’s enrolled population without any prior information about the subjects [8][11]. This technique exploits the underlying mathematical and statistical vulnerabilities of deep neural networks used in modern facial recognition systems (FRS) [8][6].

5.1.1 Construction and Physical Implementation

The primary methodology for building a Master Face is a process known as Latent Variable Evolution (LVE) [8][9]. This approach utilizes a powerful generative model, typically StyleGAN or StyleGAN2, which can synthesize high resolution, realistic human faces by manipulating latent vectors in a high dimensional space [8][9]. Because the objective function for this attack (maximizing the number of matched identities) is non-differentiable and non-convex, researchers utilize black-box optimization strategies [8][9]. The most common strategy is the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [8][9]. This algorithm iteratively searches the latent space of the generator, evolving a population of face candidates to find the specific

features that trigger a match in the largest possible part of the database [8][9]. In cases where a single face is insufficient, a greedy search algorithm is implemented to find a minimal set of faces, usually around nine, that can collectively cover more than 40 percent of a target dataset like Labeled Faces in the Wild (LFW) [8][9]. In real-world applications, these digital synthetic faces are converted into Presentation Attack Instruments (PAIs) [8][30]. Technical evaluations have demonstrated that these attacks remain highly effective when the images are printed on standard A4 paper, high-quality photo paper, or even displayed on high resolution tablets and laptop screens [8][19]. Once the physical PAI is presented to the FRS sensor, it can bypass the authentication systems used for the unlocking of mobile devices or even automated border control (ABC) systems [8][14][17].

5.1.2 Statistical Properties and Identity Space Vulnerabilities

The effectiveness of a Master Face is rooted in the non-uniform distribution of the identity embedding space [8][6][9]. FRS feature extractors are trained to map facial images into a mathematical domain where faces of the same person are clustered together and faces of different people are far apart [8][7]. However, this mapping is an approximation and results in dense areas or clusters where the discriminative power of the model is weakened [8][11]. A Master Face essentially acts as a centroid for one of these dense clusters, residing within the matching threshold of numerous disparate identities [8][11][6]. Technical analysis of these dense clusters reveals several key properties:

- **Demographic Imbalance:** Many Master Faces are statistically more likely to match demographics that are overrepresented or poorly differentiated in the training data, such as older Caucasian males [8][9].
- **Face Capacity Limits:** There is a relationship between the dimensionality of the embedding space and attack success [11]. Theoretical analysis suggests that as FRS models move toward higher dimensions and perfect identity separation, the maximum coverage of a Master Face decreases, although current practical systems still possess exploitable dense regions [11].
- **Natural Face Manifold:** Unlike simple mathematical averages of faces, Master Faces generated via LVE are constrained to the manifold of natural human faces by the GAN,

ensuring they can pass quality assessment and human scrutiny [8][6].

5.1.3 Adversarial Machine Learning Enhancements

Adversarial machine learning techniques are increasingly used to improve the potency and generalizability of Master Face attacks [8][6][7]. One significant advancement is network-assisted evolution, where a separate neural network known as a success predictor is trained to guide the evolutionary search [6][9]. This predictor estimates the probability of a match for a candidate latent vector before it is evaluated by the FRS, allowing the algorithm to explore promising areas of the latent space much more efficiently [6][9]. Attackers also utilize multi-model and multi-database optimization to create universal Master Faces [8]. By calculating similarity scores across different model architectures simultaneously (e.g., ArcFace and FaceNet), the algorithm can find features that exploit shared weaknesses in how deep neural networks perceive facial structures [8]. This creates a master "key" capable of bypassing diverse security setups, even in black-box scenarios where the attacker has no information about the target system [8][14]. Finally, these attacks are becoming more sophisticated by integrating other adversarial techniques to bypass modern defenses:

- **DeepFake Integration:** Master Faces can be animated using DeepFake technology to mimic eye blinking and mouth movements [6][9]. This allows the PAI to bypass action-based liveness detection that would normally reject static printed images [14][6][31].
- **Adversarial Noise:** Small patterns of adversarial noise can be added to the Master Face to shift its embedding specifically toward targeted clusters, combining the general effectiveness of a Master Face with the precision of a targeted evasion attack [17][32][33].
- **3D Master Faces:** New research has extended these techniques to 3D facial reconstruction [9]. By optimizing for both 2D RGB models and 3D depth sensors, these Master Faces can deceive specialized hardware found in modern high-end smartphones and border checkpoints [9].

5.2 Defenses

The literature indicates that as Master Face attacks become more sophisticated, the defensive landscape must evolve from simple one-time verification to a multi-layered approach incorporat-

ing architectural hardening, liveness detection, and multi-modal integration. This section explores the technical efficacy of these defenses and evaluates which system properties offer the highest resilience against synthetic "wolf samples."

5.2.1 Architectural Resilience

Research consistently highlights that the underlying architecture and training methodology of a Facial Recognition System (FRS) are the first line of defense against Master Face attacks. Systems that utilize angular margin loss (such as ArcFace) or uniform loss demonstrate higher resilience compared to older models like Dlib or FaceNet [8]. The core vulnerability exploited by Master Faces is the existence of dense areas or "clusters" in the embedding space where many identities overlap [8]. This is often a byproduct of unbalanced training data, particularly regarding age and gender; for instance, face embeddings for the 30–60 age group are often more scattered, while elder or younger groups may cluster, providing fertile ground for a Master Face to find matches [8][9]. To mitigate this, literature suggests two critical architectural improvements:

- **Dataset Balancing:** Robust FRS must be trained on large, balanced datasets to ensure that the feature extractor learns to distribute identities more uniformly throughout the identity space, thereby reducing the probability of a single synthetic face matching multiple templates [8].
- **High-Dimensional Embeddings:** Newer models like MagFace and ElasticFace, which often utilize higher embedding dimensions (e.g., 512 dimensions), allow for better identity separability [9]. The increased dimensionality makes the "coverage task" for a Master Face significantly more difficult because it forces the attacker to find a sample that occupies an extremely specific and isolated region of the latent space that still overlaps with others—a mathematical challenge that grows exponentially with dimension size [11].

Empirical results support this "generalizability gap." Studies show that Master Faces optimized for a specific target model show insignificant success (often <0.31%) when tested against SOTA models they were not trained on [11]. This suggests that current Master Face effectiveness is largely limited to "white-box" scenarios where the attacker has full knowledge of the target's specific embedding flaws [11][34].

5.2.2 The Role of Liveness Detection and Presentation Attack Detection (PAD)

While an FRS might be mathematically tricked into a match, Liveness Detection (LD) or Presentation Attack Detection (PAD) serves as a crucial physical gatekeeper. Literature categorizes these defenses into two primary types: Silent/Software-based and Action-based [14][30].

- **Action-based LD:** These systems require a user to perform specific involuntary or instructed responses, such as eye blinking or smiling [14]. This is considered more resilient than silent detection because it tracks the conformity of facial landmarks to expected patterns over a sequence of real-time images [14][19]. For example, the Eye Aspect Ratio (EAR) is used to ensure that a blink involves both open and closed states, a property that static Master Face images or simple replay attacks struggle to replicate.
- **Deep Learning-based Detection:** Advanced methods like Face Presentation Attack Detection analyze regional relationships and movements. For instance, models using a TimeSformer backbone can detect spoofs in as little as 8 frames (0.3 seconds) by identifying that synthetic images often exhibit more consistent directions of movement across local patches compared to bona fide faces [35].
- **GAN Artifact Detection:** Because Master Faces are typically generated using Generative Adversarial Networks (GANs), they often contain subtle digital artifacts that are invisible to the human eye but detectable by specialized GAN image or deepfake detectors [12][8].

However, critiques in the literature warn that liveness detection is not a "silver bullet." Advanced attackers can use DeepFake methods to animate a Master Face, potentially bypassing movement-based liveness checks [6][9]. Furthermore, some PAD modules are themselves vulnerable to adversarial tampering, where a mapping is learned to make "fake" class features appear as "real" distribution features [23].

5.2.3 Multi-Modal and Multi-Factor Biometric Systems

The most robust defense mentioned in the literature is the transition from unimodal (face-only) to multi-modal or multi-factor

systems. By combining face recognition with other traits like voice, fingerprints, or signatures, the system forces an attacker to satisfy multiple biometric criteria simultaneously [27][24][21].

- **Complementary Strengths:** Multi-modal systems leverage the fact that different biometrics have different failure points. For example, keystroke dynamics can complement facial recognition in low-light environments where face matching is less reliable [24].
- **Harding Against Spoofing:** A Master Face attack is essentially a "wolf sample" for a facial sensor. In a multi-modal system, even if a Master Face successfully impersonates a user's face, the attacker would still need a corresponding "master fingerprint" or "master voice" to bypass the combined check [24][28][21]. This significantly lowers the overall False Acceptance Rate (FAR) and makes the probability of a successful random impersonation virtually negligible [24][21].
- **Multi-Factor Fuzzy Extractors (MFFE):** A novel cryptographic mitigation involves embedding a secret key or password directly into the fuzzy extractor of the biometric system [36]. This prevents an attacker from reproducing authentication keys even if they possess a valid Master Face image, because the keys cannot be reconstructed without the user's private secret [36]. This provides a renewability option: if a biometric trait is compromised, the user can simply change their secret factor to render the old Master Face sample useless [36].

5.2.4 Continuous Authentication and Behavioral Monitoring

A significant depart from static entry-point security is the implementation of Continuous Authentication (CA). Unlike a standard login, CA systems quietly monitor the user throughout a session using behavioral biometrics such as gait analysis, touch dynamics, or mouse movements [27].

Dynamic Adaptability: CA systems automatically lock if they detect an identity shift or abnormal behavior [37]. This mitigates the MFA risk by ensuring that even if an attacker bypasses the initial point of entry with a Master Face, they cannot sustain access without perfectly mimicking the legitimate user's unique behavioral patterns [27].

Table 1: Summary of Master Face Mitigation Properties

Property	Description	Mitigation Strategy
Identity Separation	Forces embeddings to be further apart in the latent space to reduce identity clustering.	Angular margin loss (ArcFace) and higher-dimensional embeddings [8, 9].
Structural Integrity	Verifies that the input is a real, three-dimensional living person rather than a synthetic or printed artifact.	Action-based liveness detection (EAR/blinking) and rPPG pulse detection [14, 19].
Factor Diversity	Requires an attacker to provide multiple "wolf sample" types simultaneously.	Multi-modal fusion incorporating face, voice, and fingerprint data [24, 21, 28].
Temporal Consistency	Continuously monitors the user's identity throughout a session to detect unauthorized shifts.	Continuous Authentication (CA) using behavioral biometric tracking [27].
Cryptographic Binding	Prevents biometric data from being used to reproduce keys without a private user secret.	Multi-factor fuzzy extractors (MFFE) incorporating user-defined secrets [36].

Security-Usability Trade-off: While multi-modal and continuous systems are more secure, they can introduce latency and increased computational load, which may impact user experience on resource-constrained devices like smartphones [27][24].

5.2.5 Evaluation and Connection to Current Research

In conclusion, the literature suggests that while MFAs represent a growing threat to traditional, static FRS, they are increasingly manageable through disciplined architectural design and multi-modal integration [11][22]. The "Five C's" analysis reveals that scholars largely agree on the superiority of multi-modal systems for high-security applications, although they contrast on whether software-based detection can ever truly "out-run" GAN-based generators [8][28]. A critique of current literature is that it often ignores the survivability of Master Faces through real-world "print-scan" degradation, which may actually reduce their success rates in practical border control scenarios [34]. Our study draws upon these findings to argue that the most effective mitigation for Master Face attacks is not a single tool, but a layered security pipeline that treats facial data as only one component of a holistic identity verification process [21][5].

6 Conclusions

This study provides a comprehensive evaluation of the Master Face threat landscape, revealing that while facial recognition systems have achieved superlative performance, they remain mathematically vulnerable to synthetic "wolf samples" [23]. Our technical analysis confirms that Master Faces exploit the non-uniform distribution of identity embedding spaces, acting as centroids for dense clusters of disparate identities.

The findings rest on a critical discrepancy: early literature suggested an "extreme vulnerability" to dictionary attacks, yet our synthesis of recent research demonstrates that this vulnerability is largely limited to "white-box" scenarios where the attacker has prior knowledge of the target model's embedding flaws [6][11]. Modern FRS models, particularly those employing high-dimensional representations ($d=512$) and angular margin losses, exhibit a high degree of resilience, reducing Master Face effectiveness to levels comparable to zero-effort imposter attacks [11].

Furthermore, we identify a lack of consensus on mitigation, which remains divided between architectural fixes, such as dataset balancing to ensure uniform identity distribution and procedural defenses like GAN-artifact detectors and action-based liveness checks. We argue that the most effective mitigation is not a singular tool but a layered security pipeline. By transitioning from unimodal static verification to multi-modal systems and continuous authentication, organi-

zations can significantly lower False Acceptance Rates and render the probability of successful impersonation virtually negligible.

Ultimately, facial recognition systems are not as secure as commonly assumed. Future research must address the "digital-to-physical" transfer gap, as the survivability of Master Faces through real-world print-scan degradation remains under-explored. Advancing biometric security will require a move toward holistic evaluation protocols that test the entire pipeline; detection, liveness, and matching—simultaneously.

7 Appendix

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